Code -

```cpp
#include <iostream>
#include <vector>
#include <queue>
#include <omp.h>

using namespace std;

class Graph {
    int V;
    vector<vector<int>> adj;

public:
    Graph(int V) {
        this->V = V;
        adj.resize(V);
    }

    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    void parallelBFS(int start) {
        vector<bool> visited(V, false);
        queue<int> q;

        visited[start] = true;
        q.push(start);

        cout << "\nParallel BFS Traversal: ";

        while (!q.empty()) {
            int size = q.size();

            #pragma omp parallel for
            for (int i = 0; i < size; i++) {
                int node = -1;

                #pragma omp critical
                {
                    if (!q.empty()) {
                        node = q.front();
                        q.pop();
                        cout << node << " ";
                    }
                }

                if (node != -1) {
                    for (int neighbor : adj[node]) {
                        if (!visited[neighbor]) {
                            #pragma omp critical
                            {
                                if (!visited[neighbor]) {
                                    visited[neighbor] = true;
                                    q.push(neighbor);
                                }
                            }
                        }
                    }
                }
            }
        }
```

```
        cout << endl;
    }
};

int main() {
    int V, E;
    cout << "Enter number of vertices: ";
    cin >> V;

    Graph g(V);

    cout << "Enter number of edges: ";
    cin >> E;

    cout << "Enter edges (u v):\n";
    for (int i = 0; i < E; i++) {
        int u, v;
        cin >> u >> v;
        g.addEdge(u, v);
    }

    int start;
    cout << "Enter starting vertex: ";
    cin >> start;

    g.parallelBFS(start);

    return 0;
}
```

Output -

```
Enter number of vertices: 6
Enter number of edges: 5
Enter edges (u v):
0 1
0 2
1 3
1 4
2 5
Enter starting vertex: 0

Parallel BFS Traversal: 0 1 2 3 4 5


=== Code Execution Successful ===
```